

Type 0680L

Square Ceramic Surface Mount Slow Blow Fuse

HF 0680L Series – 2410 Size

RoHS Compliant

Features

- Slow Blow, 2410 SMD
- Compatible with 260°C, IR Pb-free solder process
- Wide range of current rating from 375mA to 12A
- Wide operating temperature range, -55°C to 125°C
- Tape & Reel for auto-insert SMD process
- RoHS compliant with exemption 7(a)
- Full compliance with EU Directive 2011/65/EU and amending directive 2015/863
- Halogen Free, (MSL = 1)
- AEC-Q Compliant
- Meets Bel automotive qualification*
- * - Largely based on internal AEC-Q test plan

Applications

- Notebook
- LCD monitor
- PC computer
- Office electronic equipment
- Industrial equipment
- Medical equipment
- POE, POE+
- LCD / LED monitor
- Power supply
- LCD / LED TV
- Storage system
- Telecom system
- Wireless basestation
- White goods
- Game console
- Battery charging circuit protection



UK
CA TÜV c UL US CE
AEC-Q Compliant

Electrical Characteristics

(UL/CSA/STD.248-14)

Testing Current	Blow Time	
	Minimum	Maximum
100%	4 Hrs.	N/A
200%	N/A	120 Sec
300%	0.15 Sec	3 Sec
800%	0.01 Sec	0.1 Sec

Safety Agency Approvals

Safety Agency	Safety Agency Certificate	Ampere Rating / Voltage Rating	Ampere Range / Volt @ I.R. ability*
	E506667	375mA-7A/125V AC 125V DC >7A-12A/50V AC 75V DC	375mA-7A/125V AC @50A 125V DC @100A 75V DC @500A >7A-12A/50V AC @100A 75V DC @500A
	50448043 EN 60127-1: 2006+A1+A2 EN 60127-4: 2005+A1+A2	375mA-7A/125V AC 125V DC >7A-12A/50V AC 75V DC	375mA-7A/125V AC @50A 125V DC @100A 75V DC @500A >7A-12A/50V AC @100A 75V DC @500A

*I.R.= Interrupting Rating = Short Circuit Rating(Amps)

Physical Specifications

Materials	Body : Ceramic
	Terminations : Silver Plated Caps /Gold Plated Caps/Palladium Plated Caps
Marking	On Fuse :
	"Current Rating", "T", "L"—laser marked on ceramic tube, "bel" stamped in end caps.
	On Label :
	"bel", "0680L", "Current Rating", "Voltage Rating", "Interrupting Rating", "Appropriate Safety Logos" and "  ", "  "(China RoHS compliant).

Specifications subject to change without notice

Environmental Specifications

Shock Resistance	MIL-STD-202G Method 213, Test condition A (50G's peak for 11 milliseconds : Half-sine waveform)	High temperature storage	MIL-STD-202 Method 108
Vibration Resistance	MIL-STD-202G, Method 201A (10-55 Hz, 0.06 inch, total excursion).	Temperature cycling	JESD22 Method JA-104, Test Condition B
Salt Spray Resistance	MIL-STD-202G, Method 101E, Test Condition B (48 hrs.).	Biased humidity	MIL-STD-202 Method 103, 85C/85% RH with 10% operating power for 1000 hrs.
Insulation Resistance	MIL-STD-202G, Method 302, Test Condition A (After Opening) 10,000 ohms minimum.	Operational life	MIL-STD-202 Method 108, Test Condition D
Thermal Shock	MIL-STD-202G, Method 107G, Test Condition B (-65°C to +125°C).	Resistance to solvents	MIL-STD-202 Method 215
Operating Temperature	-55°C to +125°C	Mechanical shock	MIL-STD-202 Method 213, Test Condition C
Moisture Sensitivity Level	1 (According to IPC J-Std-020)	Vibration	MIL-STD-202G Method 204 Test condition D (10-2k HZ , 20G's for 20minutes)
		Resistance to soldering heat	MIL-STD-202 Method 210, Test condition B
		Thermal shock	MIL-STD-202 Method 107
		Solderability	J-STD-002 Test B
		Board flex(SMD)	AEC-Q200-005
		Terminal strength	AEC-Q200-006
		Electrical characterization	3 Temperature Electrical

Electrical Specifications

Part Number	Ampere Rating	Typical Cold Resistance (ohms)	Volt-drop @100% In (Volt) max.	Voltage and Interrupting Ratings	Melting I ² T @10 In (A ² Sec)	Melting I ² T <10ms (A ² Sec)	Maximum Power Dissipation (W)	Agency Approvals	
									
0680L0375-XX	375mA	0.93	1.20	See Table of Safety Approvals on Page 1 for Voltage and associated Interrupting Ratings	0.14	0.10	0.45	Y	Y
0680L0500-XX	500mA	0.58	0.90		0.27	0.35	0.45	Y	Y
0680L0630-XX	630mA	0.41	0.80		0.43	0.42	0.50	Y	Y
0680L0750-XX	750mA	0.33	0.75		0.62	0.61	0.56	Y	Y
0680L1000-XX	1A	0.175	0.60		1.5	1.3	0.60	Y	Y
0680L1500-XX	1.5A	0.095	0.40		3.3	3.2	0.60	Y	Y
0680L2000-XX	2A	0.068	0.35		6	5	0.70	Y	Y
0680L2500-XX	2.5A	0.048	0.34		9	7	0.85	Y	Y
0680L3000-XX	3A	0.037	0.27		13	12	0.81	Y	Y
0680L3500-XX	3.5A	0.030	0.26		18	17	0.91	Y	Y
0680L4000-XX	4A	0.026	0.25		24	22	1.00	Y	Y
0680L5000-XX	5A	0.019	0.23		37	36	1.15	Y	Y
0680L6300-XX	6.3A	0.015	0.22		43	50	1.39	Y	Y
0680L7000-XX	7A	0.012	0.21		68	77	1.47	Y	Y
0680L8000-XX	8A	0.0099	0.20		108	105	1.60	Y	Y
0680L9100-XX	10A	0.0084	0.19		170	150	1.90	Y	Y
0680L9120-XX	12A	0.0063	0.18		244	180	2.16	Y	Y

Consult manufacturer for other ratings
XX - Packaging code (see "ordering information")

NOTES 1:

All tests were conducted with the fuses soldered to a printed circuit boards with a nominal thickness of 1.6 mm. The copper test circuit trace was a printed circuit with an overall length of 100 mm, copper thickness/width as described below. The printed circuit boards were mounted by screws to a test fixture having brass blocks for connection of the test leads. All samples were soldered to the test boards by the manufacturer.

Fuse rating	Test Board Trace Dimensions
375mA-5A	1 oz. copper, 5mm wide.
6A-12A	3 oz. copper, 10mm wide.

NOTES 2:

Conventional (Ambient Pressure) Reflow Process is recommended for this device. The sale and use of product is subject to bel terms and condition of sale , unless otherwise agreed . User should independently evaluate the suitability of and test each product selected for their own application. product are not designed for , and may not be used in , all applications.

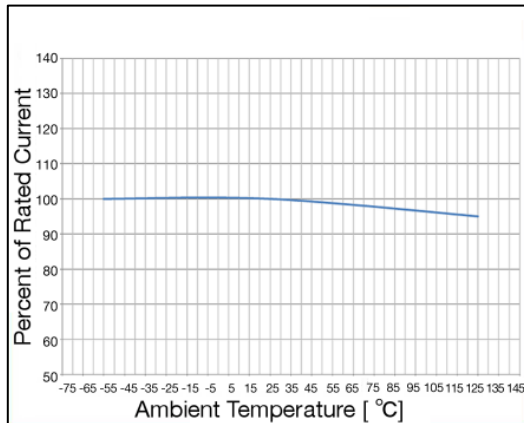


Specifications subject to change without notice

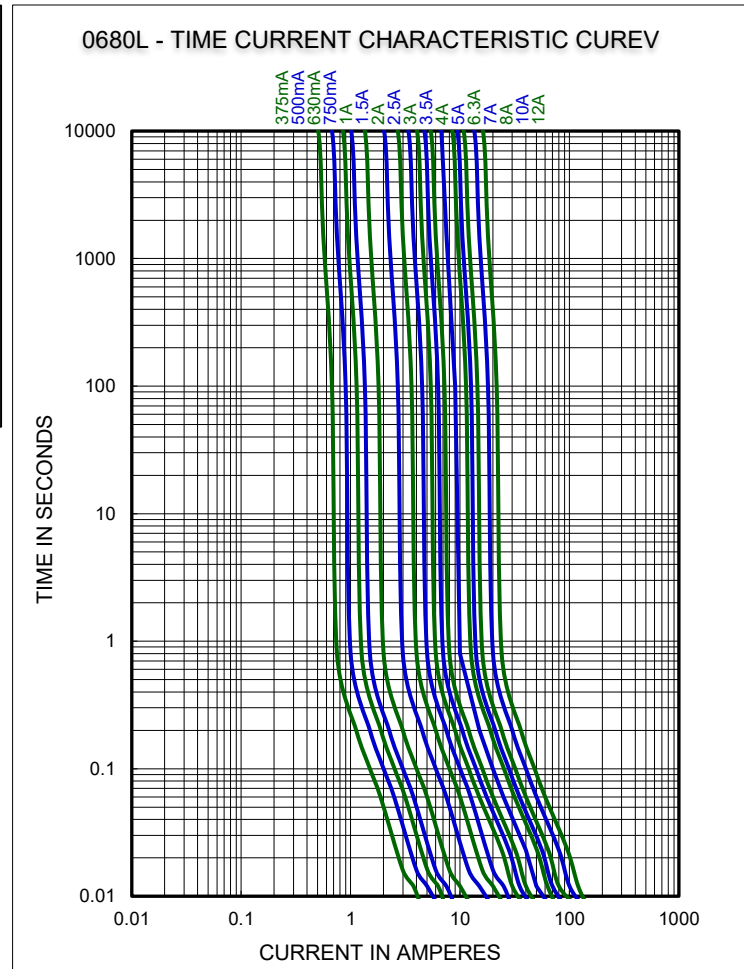
Bel Fuse Inc.
300 Executive Drive, Suite 300
West Orange, NJ 07052 USA

+1 201.432.0463
Bel.US.CS@belf.com
belfuse.com/circuit-protection

Temperature Derating Curve

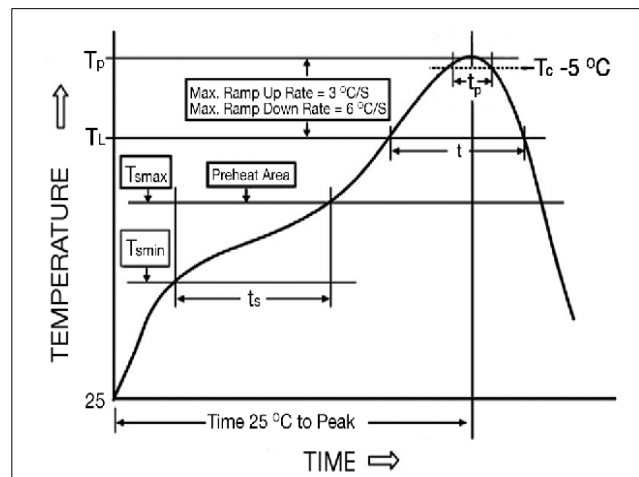


Average Time Current Curve



Soldering Parameters

IR Reflow Profile (IPC/JEDEC J-STD-020D)	
Preheat & Soak	
Temperature min (T_{smin})	150°C
Temperature max (T_{smax})	200°C
Time (T_{smin} to T_{smax}) (t_s)	60-120 seconds
Average ramp-up rate (T_{smax} to T_p)	3°C/second max.
Liquidous temperature (T_L)	217°C
Time at liquidous (t_L)	60-150 seconds
Peak temperature (T_p)	260°C max
Time (t_p) within 5°C of the specified classification temperature (T_c)	30 seconds
Average ramp-down rate (T_p to T_{smax})	6°C/second max.
Time 25°C to peak temperature	8 minutes max.



Fuse FGNO Explanation

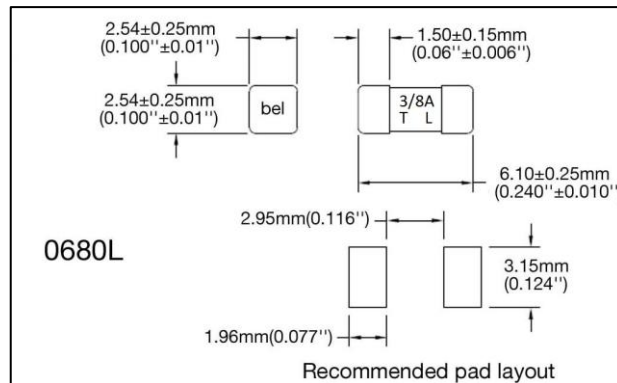
0680 L [XXXX] -XX

0680L=0680L; [XXXX]=Ampere Rating; XX=See Ordering Information as below

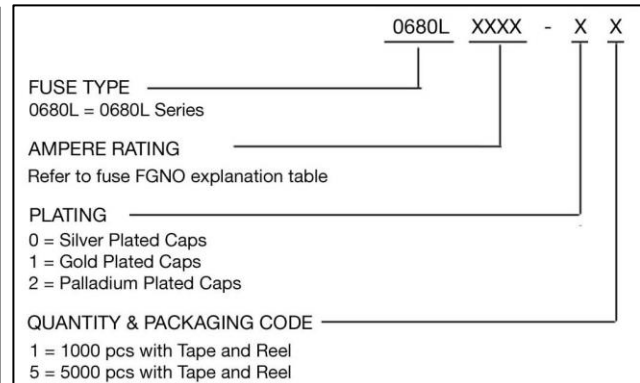
[illegible]

Fraction	Decimal	Amps	Bel FGNO[XXXX]
	1.0	1	1000
1-1/2	1.5	1.5	1500
	2.0	2	2000
2-1/2	2.5	2.5	2500
	3.0	3	3000
3-1/2	3.5	3.5	3500
	4.0	4	4000
	5.0	5	5000
	6.3	6.3	6300
	7.0	7	7000
	8.0	8	8000
		10	9100
		12	9120

Mechanical Dimensions



Ordering Information



Packaging

Packaging Tape & Reel	Packaging Specification	Quantity	Quantity & Packaging Code
12 mm wide tape with 13 inches Diameter reel	EIA Standard 481-E	5000	5
12 mm wide tape with 7 inches Diameter reel	EIA Standard 481-E	1000	1

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